

Module Checklist

Monitoring with Prometheus

By TechWorld with Nana

Video Overview



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Video Overview



Git Repos used in the module	
Prometheus Configuration Files	https://gitlab.com/twn-devops-bootcamp/latest/16-prometheus/monitoring
Nodejs Application	https://gitlab.com/twn-devops-bootcamp/latest/16-prometheus/nodejs-app-monitoring

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Introduction to Monitoring with Prometheus

- ❑ Watched video

Useful Links:

- Official Exporters List: https://prometheus.io/docs/instrumenting/exporters/
- Client Libraries: https://prometheus.io/docs/instrumenting/clientlibs/

Install Prometheus Stack in Kubernetes

- ❑ Watched video
- ❑ **Demo executed**:
  - ❑ Create EKS cluster using eksctl
  - ❑ Deploy Microservices Application
  - ❑ Explore deployed Prometheus Stack

Useful Links:

- Prometheus Helm Chart: https://github.com/prometheus-community/helm-charts

Data Visualization with Prometheus UI

- ❑ Watched video
- ❑ **Demo executed**
  - ❑ Access Prometheus UI and explore endpoints

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Introduction to Grafana

- ❏ Watched video
- ❏ **Demo executed**
  - ❏ Access Grafana
  - ❏ Test CPU Spike

Useful Links:

- Official Doc of Grafana Dashboards: https://grafana.com/docs/grafana/latest/dashboards/

Alert Rules in Prometheus

- ❏ Watched video

Useful Links:

- Alerting Overview: https://prometheus.io/docs/alerting/latest/overview/

Create own Alert Rules - Part 1

- ❏ Watched video
- ❏ **Demo executed - Create Alerting Rule Configuration**

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Create own Alert Rules - Part 2

- ❑ Watched video
- ❑ **Demo executed**
  - ❑ Created PrometheusRule component with 1st Alert Rule
  - ❑ Wrote 2nd Alert Rule
  - ❑ Applied Alert Rules

Useful Links:

- PrometheusRule Docs: https://docs.openshift.com/container-platform/4.16/rest_api/monitoring_apis/prometheusrule-monitoring-coreos-com-v1.html

Create own Alert Rules - Part 3

- ❑ Watched video
- ❑ **Demo executed - Test Alert Rule**

Introduction to Alertmanager

- ❑ Watched video

Useful Links:

- Official Doc: https://prometheus.io/docs/alerting/latest/alertmanager/
- Configuration: https://prometheus.io/docs/alerting/latest/configuration

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Configure Alertmanager with Email Receiver

- ❑ Watched video
- ❑ **Demo executed - Configure Email Notification**

Useful Links:

- AlertmanagerConfig: https://docs.openshift.com/container-platform/4.16/rest_api/monitoring_apis/servicemonitor-monitoring-coreos-com-v1.html

Trigger Alerts for Email Receiver

- ❑ Watched video
- ❑ **Demo executed - Test Email Notification**

Monitor Third-Party Applications

- ❑ Watched video

Useful Links:

- Official Docs: https://prometheus.io/docs/instrumenting/exporters/

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Deploy Redis Exporter

- ❑ Watched video
- ❑ **Demo executed**

Useful Links:

- Redis Exporter Helm Chart: https://github.com/prometheus-community/helm-charts/tree/main/charts/prometheus-redis-exporter

Alert Rules & Grafana Dashboard for Redis

- ❑ Watched video
- ❑ **Demo executed**
  - ❑ Create Prometheus Rules for Redis
  - ❑ Apply Redis Alert Rules
  - ❑ Trigger RedisDown Alert
  - ❑ Import existing Dashboard in Grafana

Useful Links:

- Collection of Prometheus Alert Rules: https://awesome-prometheus-alerts.grep.to/
- Grafana Dashboards: https://grafana.com/grafana/dashboards
- Redis Dashboard: https://grafana.com/grafana/dashboards/763

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Collect & Expose Metrics with Prometheus Client Library (Monitor own App - Part 1)

- ❑ Watched video
- ❑ **Demo executed - Deploy app into K8s cluster**
  - ❑ Built Docker Image
  - ❑ Pushed to Docker Repository
  - ❑ Created Deployment & Service Configuration
  - ❑ Created Secret component - type: docker-registry
  - ❑ Deployed App into K8s cluster

Useful Links:

- Official Client Libraries: https://prometheus.io/docs/instrumenting/clientlibs/
- Nodejs App Git Repo: https://gitlab.com/twn-devops-bootcamp/latest/16-prometheus/nodejs-app-monitoring
- Nodejs Prometheus Client:: https://www.npmjs.com/package/prom-client

Scrape Own Application Metrics and Configure Own Grafana Dashboard (Monitor own App - Part 2)

- ❑ Watched video
- ❑ **Demo executed**
  - ❑ Deployed ServiceMonitor component
  - ❑ Created Grafana Dashboard

Useful Links:

- Rate function: https://prometheus.io/docs/prometheus/latest/querying/functions/#rate